BuddyBird

Speech training for parrots

An app that plays words to a parrot while its owner is away and records the bird's attempts to imitate them. Released on Google Play and the App Store.

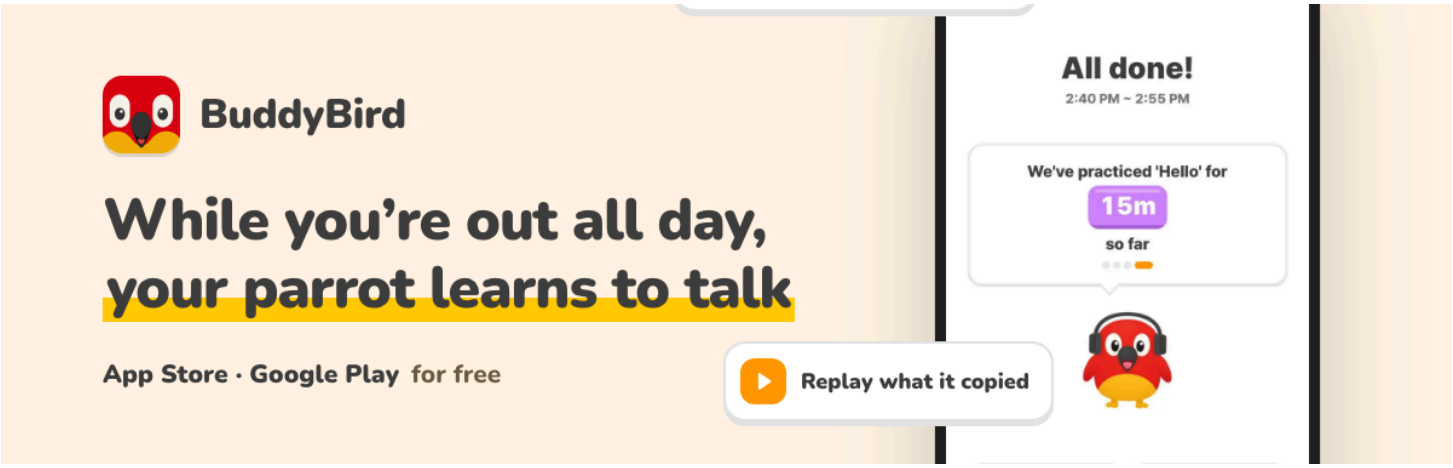
Period

Apr 2026 – Present

Role

Team lead, team of three · mobile, backend, deployment

About 1,300 cumulative downloads and 30–40 daily active users as of Oct 2026



What it does

- Repeated word playback and automatic recording of parrot speech
- Classification of parrot sounds against human speech and noise
- Remote WebRTC pet camera for monitoring training sessions
- Lock-screen controls, background playback, and custom word recordings

My contribution

- Segmented vocalizations with Silero VAD and integrated hardware AEC and WebRTC AEC3 on iOS and Android to remove playback echo.
- Built the WebRTC viewer and connected devices across networks with Cloudflare Realtime STUN/TURN.
- Integrated the parrot-sound classifier into FastAPI; moved deployment from Docker and ECS to an SQS and Lambda pipeline.
- Automated EAS and GitHub Actions builds, Maestro end-to-end tests, and mobile releases.

A problem I worked through

Problem

Words played by the app could re-enter the microphone and be mistaken for the parrot's voice.

How I solved it

I combined vocalization detection with native echo cancellation to separate playback from recorded speech.

Process and decisions

From the problem to an MVP

The midterm review framed the gap: video loops and generic pet cameras could not play an owner's voice or teach selected words. We chose voice-based word training as the first feature and launched the app on June 10, 2026.

Consented audio collection

In the initial MVP, VAD isolated intervals from training recordings collected with user consent. Audio went to S3 and metadata such as duration and format to RDS. By Aug 14, the pipeline had collected 11,250 clips totaling about 1.87 GB for model validation.

Evolving the shipped product

Remote pet monitoring and training results grew from the review roadmap into product features. To prevent playback audio from confusing vocalization detection, I integrated Silero VAD and platform-specific echo cancellation into the audio path.

What the midterm review validated

As presented in Aug 2026

The MVP launched on June 10 to test user demand and whether parrot vocalizations could be collected for later analysis.

824
Cumulative downloads by Aug 15

~32
Daily active users, Jul 1–Aug 15

11,250
Audio clips collected by Aug 14

Audio collection and classification

01 Record during training with user consent

02 Isolate sound intervals with VAD

03 Store audio in S3 and metadata in RDS

04 Classify parrot sounds using Perch 2.0 features and logistic regression

We selected Perch 2.0 after Leave-One-Bird-Out validation across conures and quakers. The early DTW subsequence prototype for word imitation reached 0.83 AUC and 0.77 balanced accuracy. These are research results from the midterm review.

Emergency detection and daily activity reports were presented as future work at that stage.

Technology

React Native (Expo) **WebRTC** **Python · FastAPI** **AWS ECS · SQS · Lambda** **Docker · Pulumi**

EAS · GitHub Actions · Maestro